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ORIGINAL ARTICLE

Efficacy of home-based low-level laser therapy in patients with low back pain: a double-blinded randomized controlled trial

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Abstract

This study tests the effect of home-based portable low-level laser therapy(spot size of the laser 0.16 cm2, consisting of a total 110 pieces of multi-wavelength vertical-cavity surface-emittinglaser diodes(55 units at 670nm with 1mW output each, 25 units at 780nm with 2mW output each, 10 units at 830nm with 15mW output each, and 20 units at 910nm with 5mW output each) for treatment of patients with chronic low back pain. Thirty patients with low back pain were enrolled in this prospective, randomized, sham-controlled, home-based self-therapy trial. The experimental group (n = 15) received laser stimulation of the painful area of their backs using the CarerayTM, while the control group (n = 15) received sham stimulation using identical equipment that generated a red light only without laser output. Both groups self-treated for 20 minutes a day, 5 days a week for 3 weeks (a total of 15 sessions).

The primary outcome was assessed using a visual analogue scale (VAS) for low back pain, ranging from 0 (painless) to 10 (extreme pain). The secondary outcomes were measured using the Oswestry Disability Index-Korean test (ODI-K), Schober's test, and a 36-item Short Form Survey (SF-36). Outcome measurements were taken before and after treatment. All thirty participants provided outcome data. After the intervention, both groups reported significantly reduced VAS scores. In the experimental group, scores decreased from a pre-treatment average of 4.38±1.19 to a post-treatment average of 3.25±1.04, a change of Δ1.13±1.13 (p = 0.026). The control group's scores dropped from a pre-treatment average of 4.00±0.53 to a post-treatment average of 2.88±0.83, a change of Δ1.13±0.83 (p = 0.024) in the within-group comparison. Both group showed a significant improvement in Schober's test results (experiment group: p = 0.007, control group: p = 0.002). In the ODI-K and SF-36, both group showed improvements, but only the control group showed significant improvements. There were no significant differences between the two groups across all outcomes, and no side effects were reported. Medical laser irradiation is a portable and easy to use laser therapeutic modality without side effects. Further research is needed regarding physical therapeutic modalities that are readily accessible to patients experiencing chronic low back pain.

Keywords Laser therapy · Low back pain · Quality of life · Physical therapy modalities · Photobiomodulation

Introduction

Low back pain (LBP) is one of the most common musculoskeletal disorders [1, 2] and the leading cause of disability worldwide [3]. In 2020, low back pain affected 619 million people globally, with a projection of 843 million prevalent cases by 2050 [4]. Musculoskeletal disorders account for 6–8% of total disability-adjusted life years, of which LBP represents nearly half [5].

There are various treatment modalities for LBP in clinical and home environments. In a hospital setting, non-pharmacological therapies (such as exercise, cognitive behavioral

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therapy, spinal manipulation, massage, and acupuncture), pharmacological therapies, interventional therapies (such as spinal epidural injections, facet joint injections), and surgery are treatment options [6–8]. At home, postural education, exercise education, superficial heat, and mindfulness-based stress reduction can be tried [6].

In recent years, low-level laser therapy (LLLT) has emerged as a potential therapeutic strategy, with several reports suggesting significant reductions in LBP-related pain [8]. LLLT has been used since the mid-1960s for conditions requiring stimulation of healing, relief of pain and inflammation, and functional restoration [9]. This therapy utilizes non-ionizing light, including low-level lasers and light-emitting diodes, to modulate cellular activity.

LLLT is thought to provide analgesic, myorelaxant, tissue-healing, and biostimulatory effects that are beneficial for musculoskeletal disorders [10–12]. Importantly, it is considered safe due to its low power output, minimizing the risk of burns, tissue damage, or symptom exacerbation. Unlike high-power lasers, LLLT produces minimal heat, reducing the chance of thermal injury [13]. These safety features make it suitable for home use with minimal adverse effects.

Many systemic reviews and randomized controlled studies have demonstrated the effectiveness of LLLT in reducing pain and disability [8, 11, 14]. Traditionally, LLLT devices are large heavy medical laser irradiators used in clinical contexts. Careray™ is a newly innovated portable laser irradiator designed for easy home-based treatment and approved for this purpose by the Korean Ministry of Food and Drug Safety. To date, no studies have investigated the use of LLLT devices for treating LBP at home. Therefore, we examined the therapeutic efficacy and potential side effects of this home-based portable laser irradiator (Careray™) in patients with LBP.

Methods

Study design

This study was a prospective, randomized, sham-controlled, home-based trial, registered with the Clinical Research Information Service, which is under the authority of the Korea Disease Control and Prevention Agency (Registration number: KCT0008106). The Institutional Review Board approved the study (IRB No. 2020-07-061).

Study protocol

Both groups were self-treated at home for a total of 15 sessions: 20 min per day, 5 days per week, over the course of 3 weeks. The experiment group was treated with the medical laser irradiator while the control group received sham stimulation that mimicked the irradiator's shape and color. The clinical research coordinator instructed the patients to keep a daily treatment diary. Participants were permitted to take medication they had started at least a month before screening. They were not allowed to change the dose or type of medication or start any other types of treatments for LBP during the trial. Patients were assessed twice: on the day before treatment began (pre-treatment) and again within four days of the end of treatment (post-treatment). Each patient completed a daily self-therapy checklist to track their treatment and record any adverse effects.

Randomization

Study participants were randomized into either the experimental group (treated with the Careray™) or the control group. Simple randomization was done using the random number table found at Randomization.com. The seed values used for randomization were also recorded. Randomization was completed before the first participant was screened, and the results were generated by a researcher who was not involved in the study enrollment or intervention. The assessors collecting study data were unaware of the group assignments throughout the trial.

Participants

This study was conducted at OO University Hospital's rehabilitation center between October 2021 and October 2022. Study participants were recruited through a notice posted on a bulletin board within the hospital and screened by a rehabilitation physician. Patients were selected based on the following inclusion criteria: (1) adult men and women aged 19 to 70 years, (2) individuals experiencing chronic pain in the lumbar–sacral region for 12 weeks or more, (3) persons possessing sufficient cognitive abilities to accurately describe the location and intensity of their pain, and (4) individuals who fully understood the research's purpose and procedures and wished to participate voluntarily. The exclusion criteria were (1) individuals who had received lumbar injections or undergone spinal surgery within the last month, (2) people with back pain due to infectious spondylitis, fractures, or tumors, (3) individuals with a history of sensitivity to lasers, (4) individuals with mental disorder, (5) Individuals who are pregnant or planning to become pregnant, (6) Others deemed unsuitable for participation in the clinical study by the researcher for various reasons.

Each participant who fulfilled the inclusion criteria was randomly assigned to one of the two groups by a clinical research coordinator (a clinical research nurse not involved in the assessments). The assignments were concealed from

the investigators. Participants were shown the treatment procedure and warned of potential side effects. Written informed consent was obtained from all participants before randomization. All research was conducted in accordance with the ethical standards outlined in the Declaration of Helsinki [15].

Intervention

The Careray™ (WellsCare Co., Seoul, Korea) used in this study is a lightweight, wearable, and flexible LLLT device (415 g) designed for home therapy. The device incorporates multi-wavelength vertical-cavity surface-emitting laser diodes (670, 780, 830, and 910 nm), with a total treatment area of 92.7 cm^2. During a 20-minute session, the device delivered an average power density of 3.83 mW/cm^2 and an average energy density of 4.59 J/cm^2, with pulsed emission at 625 Hz and an estimated penetration depth of 2–5 mm. The device was attached to the lower back area where each patient reported pain using an elastic band (Fig. 1). Detailed technical specifications, including diode configuration, spot size, and per-diode energy calculations, are provided in the Supplementary Material (Table S1).

For the sham therapy, an identical-looking device with the same configuration was used, except that it contained low-power LEDs (650 nm, 0.02 mW per diode) instead of lasers. This ensured visual mimicry while preventing therapeutic laser energy delivery.

Outcome measures

All measurements were captured by researchers who were blinded to the treatment allocation and otherwise uninvolved in the intervention.

Primary outcome

The primary outcome was assessed using a visual analogue scale (VAS) for LBP. Each patient was instructed to mark their pain on a 10-cm-long scale according to severity. A score of 0 indicated no pain while 10 indicated extreme pain [16].

Secondary outcomes

Secondary outcomes were measured using the Oswestry Disability Index-Korean test (ODI-K), Schober's test, and a 36-item Short Form 36 (SF-36).

The ODI-K is a valid and reliable instrument used to assess the degree of disability in LBP patients. It comprises 10 items, namely, pain, sleeping, walking, carrying items, personal care, standing, sitting, social and sexual activities, and traveling. Each item is scaled from 0 to 5, so total scores range from 0 to 50, with higher values representing greater disability [17, 18].

In Schober's test, the lumbosacral junction (as indicated by the dimples of Venus) was marked while the participant stood erect, and a second mark was placed 10 cm above the junction. The participant was then asked to bend forward as far as possible as though to touch the toes, and the new distance between the marks was measured [19].

The participants' health-related quality of life was assessed through the Medical Outcomes Study 36-Item Short Form Health Survey (SF-36). The 8 components (physical functioning, physical role functioning, bodily pain, general health perceptions, vitality, social role functioning, emotional role functioning, and mental health) were aggregated into a physical component summary and a mental component summary. Scores on each subscale ranged from 0 to 100, with higher scores indicating better health. A difference of more than 2 points was considered clinically meaningful for both the physical and mental subscales [20].

Sample size

In a previous study conducted by Djavid et al., the efficacy of LLLT combined with exercise was compared to

Fig. 1 Features of Careray™



sham laser therapy plus exercise on Visual Analogue Scale (VAS) scores after 12 weeks. In their study, the difference in VAS scores between the experimental group and the control group was calculated with a mean of 1.8 and a standard deviation of 1.503 when assessing the change in VAS scores between baseline and 12 weeks post-treatment. Therefore, based on an effect size of 1.8, a standard deviation of 1.503, a significance level of 0.5, a power of 0.8, and a dropout rate of 20%, the required minimum sample size for this study was calculated to be 14 participants per group, resulting in a total of 28 participants. Ultimately, we enrolled 30 patients.

Statistical analysis

Statistical analysis was performed using SPSS 23.0 software for Windows (SSPS Inc., Chicago, IL, USA). Data were presented as mean (SD) for continuous variables and frequency for categorical variables. In the analysis of baseline demographics, we conducted the Pearson chi-squared test for categorical variables, the independent t-test for numerical variables assuming normality, and the Mann-Whitney U test for numerical variables when normality assumptions were not met. Within-group comparisons were assessed by paired t-test when the assumption of normality was satisfied and the Wilcoxon signed-rank test was used otherwise. For comparisons between groups, the independent t-test was used when the assumption of normality was satisfied and the Mann–Whitney U-test was used otherwise. Effect sizes (Cohen's d) were calculated for between-group comparisons of primary and secondary outcomes to assess clinical relevance. A post hoc power analysis based on the observed effect size in VAS was also conducted using G*Power 3.1.9.7. A p-value$\leq$0.05 was considered statistically significant.

Results

Thirty patients were recruited between 2 October 2021 and 17 August 2022 and randomly allocated to the experimental (n=15) or control group (n=15). All participants completed treatment and evaluation (Fig. 2). Baseline demographic and clinical characteristics, including age, gender, and pre-treatment outcome scores (VAS, ODI-K, Schober's test, SF-36), did not differ significantly between groups (Table 1).

Effect of intervention

Primary outcomes

Both groups demonstrated significant reductions in VAS scores after treatment (experimental: mean reduction −0.8; control: −0.9). However, the magnitude of improvement did not differ significantly between groups, and the between-group effect size was very small (Cohen's d=0.080). Detailed values are provided in Table 2.

Secondary outcomes

Both groups showed improvements in secondary outcomes, including ODI-K, SF-36, and Schober's test. For example, disability scores (ODI-K) decreased and functional mobility (Schober's test) improved modestly in both groups, while quality of life scores (SF-36) also showed upward trends. However, none of these

Adverse events

No adverse events or safety concerns were reported during the trial.

Discussion

In this study, we evaluated the efficacy of home-based LLLT for chronic low back pain. Both the active and sham groups showed significant improvements in VAS and secondary outcomes; however, no significant between-group differences were detected. The small effect size (Cohen's d=+0.076 for VAS) suggests minimal additional benefit of active treatment beyond sham stimulation. Secondary outcomes, including ODI-K, SF-36, and Schober's test, also improved within both groups without significant between-group differences, consistent with the primary outcome.

LLLT employs low-powered light (1–1000 mW, 632–1064 nm) to induce photobiomodulation, promoting anti-inflammatory, analgesic, and tissue-repair processes through mitochondrial activation and modulation of inflammatory mediators [10–13, 19, 21]. Based on these mechanisms, we expected our device to reduce inflammation and alleviate pain. Prior studies reported reductions in PGE2, TNF-α, and COX-2, along with improved microcirculation and ATP production [22–32]. Nevertheless, the limited penetration depth (estimated 1–5 mm) of our device may not have been sufficient to affect deeper structures implicated in CLBP, such as intervertebral discs, facet joints, and nerve roots [33]. Central sensitization may also contribute to persistent symptoms [34, 35].

Evidence suggests that LLLT is more effective in conditions involving superficial tissues, such as wrist and neck pain [36, 37]. In low back pain, outcomes appear to depend on both dosage and device characteristics. A meta-analysis by Glazov et al. highlighted stronger effects with higher doses (> 3 J) and shorter pain duration (< 30 months) [14]. Studies by Konstantinovic et al. and Vallone et al. also reported positive outcomes when larger spot sizes (1.0 cm² and 32 cm², respectively) and higher energy delivery were used [38, 39]. By comparison, our

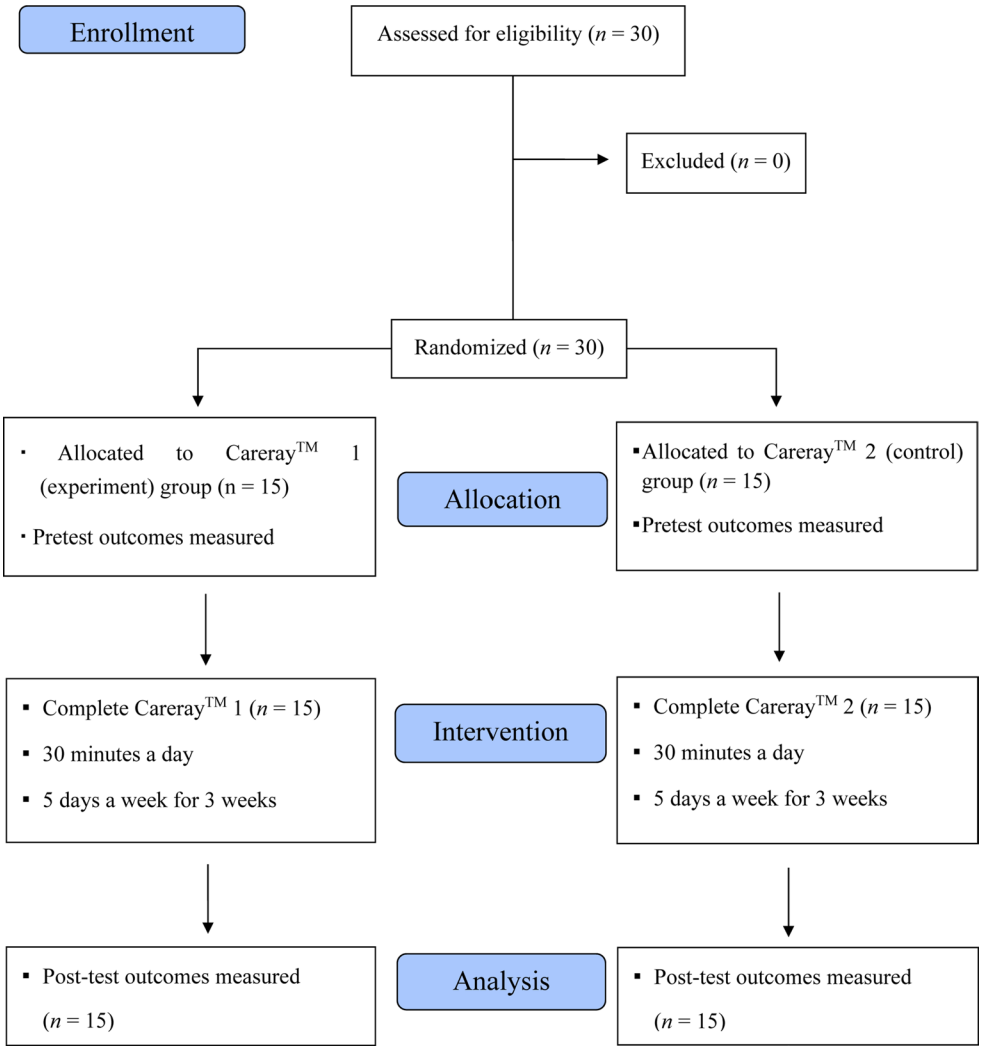


Fig. 2 CONSORT flow diagram

Table 1 Baseline demographic characteristics.

	Control group ($n=15$)	Experiment group ($n=15$)	p-value
Age	48.07 ± 12.78	46.60 ± 12.61	0.754[c]
Sex (male: female)	3:5	5:3	0.754[b]
VAS	4.00 ± 0.53	4.20 ± 1.08	0.689[c]
ODI-K	12.33 ± 3.68	12.13 ± 4.73	0.402 [c]
Schober's test	5.18 ± 0.61	5.71 ± 1.46	0.621 [c]
SF-36	109.47 ± 13.10	110.07 ± 13.27	0.152 [a]

[a]independent t-test, [b]pearson Chi-square, [c]Mann–Whitney U test; *VAS* visual analogue scale, *ROM* range of motion, 36-Item short form health survey: SF-36

device operated within WALT guidelines (4–8 J/cm²) but with relatively low power density (3.83 mW/cm²) and small spot size (0.16 cm²), limiting energy delivery per point [40]. The biphasic dose-response phenomenon (Arndt–Schulz Law) further suggests that insufficient dosing may result in suboptimal outcomes [41]. As a home-use device, its design prioritized safety through lower power output and longer treatment times, which may have reduced efficacy in deeper tissues.

Placebo effects may also explain part of the improvement. The sham device emitted visible LED light (650 nm, 0.02 mW), which, combined with patient expectations and rest, could contribute to perceived benefits. To better distinguish true therapeutic effects from placebo, future studies should incorporate objective biomarkers (e.g., C-reactive protein, cytokines) and imaging (e.g., MRI) to assess biological and structural responses.

Post-hoc power analysis indicated that the study may have been underpowered to detect small but clinically relevant effects. While originally powered for a large effect (ΔVAS = 1.8), the observed difference was < 1.0. Assuming ΔVAS = 1.0 (SD = 1.0), power was ~ 84%, but fell to ~ 62% for ΔVAS = 0.8. Thus, subtle treatment effects may not have been detected.

This study has limitations, including limited penetration depth and power output of the device, small sample size, absence of subgroup analyses, short follow-up, and lack of objective biological measures.

In conclusion, this home-based LLLT device was safe and convenient but did not demonstrate superiority over

Table 2 Comparison of the primary and secondary outcomes in LLLT and control group pre- and post-treatment

Outcomes	V1	V2	V2 − V1	Intra *p* value	Inter *p* value	Effect size
VAS						
LLLT group	4.20 ± 1.08	3.40 ± 0.83	−0.80 ± 1.01	0.009[*b]	0.689[d]	0.080
Control group	4.00 ± 0.53	3.13 ± 0.99	−0.87 ± 0.83	0.006[*a]		
ODI-K						
LLLT group	12.13 ± 4.73	10.60 ± 3.85	−1.53 ± 4.39	0.198[b]	0.402[d]	0.528
Control group	12.33 ± 3.68	8.53 ± 4.34	−3.80 ± 4.20	0.006[*a]		
SF-36						
LLLT group	110.07 ± 13.27	112.60 ± 13.76	2.53 ± 7.04	0.158[a]	0.152[c]	−0.538
Control group	109.47 ± 13.10	118.27 ± 13.43	8.80 ± 14.90	0.024[*a]		
Schober's test						
LLLT group	5.71 ± 1.46	5.94 ± 1.41	0.23 ± 0.28	0.007[*a]	0.621[d]	−0.344
Control group	5.18 ± 0.61	5.61 ± 1.12	0.43 ± 0.77	0.002[*a]		

Values are given as means ± SD

[a] Wilcoxon signed-rank test. [b] Paired t-test. [c] Independent t-test, [d] Mann–Whitney U test

V1, pre-treatment; V2, post-treatment

Effect sizes (Cohen's d) represent between-group comparisons of pre-post change scores (Δ values)

sham treatment in CLBP. Optimizing device parameters (power output, spot size, dosage), incorporating biomarkers and imaging, and stratifying patient subgroups may enhance the therapeutic potential of portable LLLT systems in chronic musculoskeletal conditions.

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Data availability The datasets generated and/or analyzed during the current study are included in this article. Further inquiries can be directed to the corresponding author, and additional data are available upon reasonable request.

Declarations

Ethics approval This study was approved by the Institutional Review Board of Jeonbuk National University Hospital (Approval number: 2020-07-061). Study participants were recruited through a notice posted on a bulletin board within the hospital and their suitability for inclusion was screened by a rehabilitation physician. A written declaration of informed consent was signed by all the patients before randomization. The trial was conducted in accordance with the Declaration of Helsinki.

Conflict of interest The authors declare no competing interests.

Clinical trial registration number KCT0008106.

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